

```
In [2]: import numpy as np
import pandas as pd
import seaborn as sns
```

```
In [3]: df=pd.read_csv('heart.csv')
```

```
In [4]: df.head()
```

```
Out[4]:
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	0
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	0
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	0



```
In [5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1025 non-null   int64
1   sex         1025 non-null   int64
2   cp          1025 non-null   int64
3   trestbps    1025 non-null   int64
4   chol        1025 non-null   int64
5   fbs         1025 non-null   int64
6   restecg     1025 non-null   int64
7   thalach     1025 non-null   int64
8   exang       1025 non-null   int64
9   oldpeak     1025 non-null   float64
10  slope       1025 non-null   int64
11  ca          1025 non-null   int64
12  thal        1025 non-null   int64
13  target      1025 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
```

```
In [7]: df.shape
```

```
Out[7]: (1025, 14)
```

```
In [8]: df.describe()
```

Out[8]:

	age	sex	cp	trestbps	chol	fbs	restecg
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	54.434146	0.695610	0.942439	131.611707	246.000000	0.149268	0.500000
std	9.072290	0.460373	1.029641	17.516718	51.59251	0.356527	0.500000
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000
25%	48.000000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000
50%	56.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	275.000000	0.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000

In [16]:

```
A=df.dropna()
A
```

Out[16]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2
...	...	...	...	...	...	...	...	...	...	...	...	...	...
1020	59	1	1	140	221	0	1	164	1	0.0	2	0	2
1021	60	1	0	125	258	0	0	141	1	2.8	1	1	3
1022	47	1	0	110	275	0	0	118	1	1.0	1	1	2
1023	50	0	0	110	254	0	0	159	0	0.0	2	0	2
1024	54	1	0	120	188	0	1	113	0	1.4	1	1	3

1025 rows × 14 columns

In [18]:

```
df.isnull()
```

Out[18]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1020	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1021	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1022	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1023	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1024	False	False	False	False	False	False	False	False	False	False	False	False	False	False

1025 rows × 14 columns



In [17]: `df.isnull().sum()`

```
Out[17]: age          0
sex          0
cp           0
trestbps     0
chol         0
fbs          0
restecg      0
thalach      0
exang        0
oldpeak      0
slope        0
ca           0
thal         0
target       0
dtype: int64
```

In [19]: `df_cleaned = df.dropna()`

```
print("\nData after dropping rows with null values:")
print(df_cleaned)
```

Data after dropping rows with null values:

	age	sex	cp	trestbps	chol	fb	restecg	thalach	exang	oldpeak	\
0	52	1	0	125	212	0	1	168	0	1.0	
1	53	1	0	140	203	1	0	155	1	3.1	
2	70	1	0	145	174	0	1	125	1	2.6	
3	61	1	0	148	203	0	1	161	0	0.0	
4	62	0	0	138	294	1	1	106	0	1.9	
...	...	...	..	...	...	...	...	...	...	...	
1020	59	1	1	140	221	0	1	164	1	0.0	
1021	60	1	0	125	258	0	0	141	1	2.8	
1022	47	1	0	110	275	0	0	118	1	1.0	
1023	50	0	0	110	254	0	0	159	0	0.0	
1024	54	1	0	120	188	0	1	113	0	1.4	

	slope	ca	thal	target
0	2	2	3	0
1	0	0	3	0
2	0	0	3	0
3	2	1	3	0
4	1	3	2	0
...	...	..	...	...
1020	2	0	2	1
1021	1	1	3	0
1022	1	1	2	0
1023	2	0	2	1
1024	1	1	3	0

[1025 rows x 14 columns]

```
In [30]: #Model building
from sklearn.model_selection import train_test_split
from sklearn import linear_model , metrics
```

```
In [31]: x=df[['age']]
y=df[['sex']]
```

```
In [32]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
In [33]: len(x_train)
```

```
Out[33]: 820
```

```
In [41]: len(x_test)
```

```
Out[41]: 205
```

```
In [39]: len(y_train)
```

```
Out[39]: 820
```

```
In [40]: len(y_test)
```

```
Out[40]: 205
```

```
In [42]: reg=linear_model.LinearRegression()
```

```
In [43]: print(x_train)
```

```
      age
835    49
137    64
534    54
495    59
244    51
..     ...
700    41
71     61
106    51
270    43
860    52
```

```
[820 rows x 1 columns]
```

```
In [44]: model=reg.fit(x_train,y_train)
```

```
In [45]: r_sq=reg.score(x_train,y_train)
```

```
In [46]: print("determination coeff",r_sq)
```

```
determination coeff 0.0069841902799135935
```

```
In [47]: print("intercept",model.intercept_)
```

```
intercept [0.92727167]
```

```
In [50]: print("slope",model.coef_)
```

```
slope [[-0.00418032]]
```

```
In [53]: y_predict = model.predict(x_test)
y_predict
```

```
Out[53]: array([[0.66809194],
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